

# Cybersecurity Incident Report

## Section 1: Identify the type of attack that may have caused this network interruption

One potential explanation for the website's connection timeout error is a DoS attack, specifically a SYN flood.

The logs show repeated SYN requests from the source IP address 203.0.113.0. These connection attempts are not consistently completed through the full TCP three-way handshake. This pattern can cause the server to maintain multiple half-open connections and consume resources that would otherwise be available to legitimate users.

## Section 2: Explain how the attack is causing the website to malfunction

When website visitors establish a TCP connection with the web server, a three-way handshake occurs:

1. The client sends a SYN packet to the server to request a connection.
2. The server responds with a SYN-ACK packet and temporarily reserves resources for the connection.
3. The client sends a final ACK packet to the server, completing the handshake.

During a SYN flood attack, a malicious actor sends a large number of SYN packets without completing the final step of the handshake. This causes the server to maintain many half-open connections and gradually consume its available connection resources.

The logs indicate repeated connection attempts from 203.0.113.0, while some legitimate connection attempts are unable to complete successfully. As the server becomes overwhelmed, legitimate users may experience connection timeouts or rejected connections because the server has insufficient resources to process new requests.

To help mitigate this type of attack, the organization can implement SYN cookies to reduce resource usage from half-open connections and apply rate limiting or firewall/IPS rules to restrict excessive SYN requests.